Adaptive Hybrid Semantic Research Assistant: Retrieval-First, Evidence-Grounded Extractive Question Answering

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***Abstract*—Keyword search misses paraphrases, dense semantic search misses exact terms, and generative language models can produce fluent answers that no source supports. This paper presents the Adaptive Hybrid Semantic Research Assistant, a retrieval-first question answering system in which every answer is a span copied from a retrieved passage. The system fuses BM25 and Sentence-BERT retrieval with Reciprocal Rank Fusion, expands each question into rule-based variants, re-ranks candidates with a cross-encoder, and extracts the answer with a fine-tuned transformer reader. On SQuAD v1.1 the retrieval stack raises Recall@10 from 86.15% for BM25 to 96.25% and Mean Reciprocal Rank from 69.49% to 85.25%, removing 73% of first-stage retrieval misses. The RoBERTa-base reader reaches 85.38 Exact Match and 91.74 F1 on the full 10,570-question development set, which exceeds human performance, the published BERT-base and DistilBERT results, and the published single-model BERT-large result, and is within 0.1 F1 of the BERT-large ensemble. End to end, the system answers 71.0% of questions exactly. The whole stack uses open models, runs on one 8 GB laptop GPU, and returns evidence, confidence and latency with every answer.**

***Index Terms*—question answering, hybrid retrieval, reciprocal rank fusion, cross-encoder re-ranking, extractive reading comprehension, SQuAD, RoBERTa**

I. Introduction

Open-domain textual question answering (QA) answers a question from a large collection instead of from one supplied paragraph. Progress in the task is driven by two developments, machine reading comprehension and neural information retrieval, which together let a system retrieve relevant text and read it to extract an answer [1]. The retrieval side is commonly organised in two stages: a fast first stage selects candidates from the whole collection, and a slower, stronger second stage re-scores them [2].

Each building block has a known weakness. Lexical ranking with BM25 [3] rewards exact matches of distinctive terms but gives no credit to a passage that answers the question in different words. Dense retrieval with sentence embeddings [4] matches meaning but can under-weight rare names and identifiers. Generative answering reads well, yet unfaithful output is a major risk wherever language models would otherwise be useful [5].

This paper describes a system that combines the complementary strengths and removes the faithfulness risk by construction. It retrieves with both BM25 and a bi-encoder, fuses the rankings by position, re-ranks the fused candidates with a cross-encoder, and extracts the answer as a span of the top passage. An answer therefore cannot contain text that is absent from the evidence shown to the user.

The contributions are:

- a complete hybrid retrieval and extractive QA pipeline built only from open models that run locally on a single 8 GB laptop GPU;
- a lightweight adaptive query layer, with rule-based complexity estimation that sets the retrieval depth and multi-query expansion that needs no additional model;
- a seven-configuration ablation that measures the contribution of every retrieval stage with standard ranking metrics on a public benchmark;
- a dual-setting evaluation of three readers, with the gold passage and end to end behind the same retriever, which locates the remaining error in the pipeline;
- results that match or exceed published reference points: 96.25% Recall@10 for retrieval and 85.38 EM / 91.74 F1 for reading on SQuAD v1.1.

II. Related Work

A. Sparse and Passage-Level Retrieval

BM25 scores a passage by summing, over the query terms, an inverse-document-frequency weight multiplied by a saturating term-frequency component with length normalisation [3]. It needs no training data. Dang *et al.* [6] showed on several TREC collections that scoring documents through their passages performs significantly better than scoring whole documents, which supports indexing at a finer granularity than full articles.

B. Dense Retrieval and Vector Search

Text representation has moved from rule-based and statistical methods to contextual learned embeddings [7]. Sentence-BERT encodes each sentence independently into a fixed vector, reducing the search for the most similar pair among 10,000 sentences from about 65 hours with BERT to about 5 seconds [4]. Dual encoders are now widely used for open-domain QA: Dense Passage Retrieval trains them with in-batch negatives [8], and adaptive batch scheduling makes those negatives progressively harder [9]. Tan *et al.* [10] observe that embedding a long text into one vector loses information, a direct motivation for chunking. FAISS provides exact and approximate nearest-neighbour search at billion scale [11].

C. Hybrid Retrieval, Fusion and Query Expansion

BM25 scores and cosine similarities lie on unrelated scales. Reciprocal Rank Fusion (RRF) avoids calibration by combining rankings by position, and outperformed Condorcet fusion and individual learning-to-rank methods [12]. Recent IEEE work confirms the value of hybrids: Bindle *et al.* [13] combine TF-IDF and BM25 with Sentence-BERT and Word2Vec using query-adaptive weights, and Khan *et al.* [14] fuse BM25 with dense embeddings through RRF inside a retrieval-augmented generation system. Query expansion addresses the mismatch between one phrasing of a question and the wording of the answering passage; Raza *et al.* [15] organise semantic expansion methods into a taxonomy, and Mageshwari and Shrijith [16] combine language-model expansion with RRF.

D. Neural Re-ranking

Keyword features cannot capture contextual relationships between query and passage, whereas BERT-based rankers capture them far better at the cost of labelled training data [17]. Kim and Ko obtained the best non-ensemble MRR@10 on the MS MARCO passage re-ranking leaderboard with multitask fine-tuning [17].

E. Extractive Reading Comprehension

SQuAD contains over 100,000 crowd-sourced questions whose answers are spans of Wikipedia paragraphs [18]. BERT introduced deep bidirectional pre-training followed by fine-tuning [19]; RoBERTa showed that BERT was under-trained and improved the pre-training recipe [20]; DistilBERT is 40% smaller and 60% faster while retaining 97% of BERT's language-understanding capability [21]. Liu *et al.* [22] note that retrieve-then-extract pipelines suffer from error propagation. Sinthusha *et al.* [23] compare transformer readers on a translated SQuAD, and Huang *et al.* [24] show that distillation retains over 95% of BERT's accuracy with about half the parameters.

F. Grounding and Faithfulness

Hypothetical Prompt Embeddings improve chunk-level retrieval for grounded answers [25], and Malin *et al.* [5] review how faithfulness is measured across summarisation, QA and translation.

G. Positioning

Table I places this work against the six most closely related IEEE publications. Earlier systems tend to report end results with generation-oriented metrics or to depend on proprietary or per-query language models. This work keeps the same proven building blocks, runs them with open models only, and reports the contribution of every stage with ranking metrics.

III. System Architecture

Fig. 1 shows the complete system. An offline pipeline builds the indexes once; an online pipeline answers each question in eight steps.

A. Corpus Construction and Adaptive Chunking

All contexts of the SQuAD v1.1 training and development sets are de-duplicated by exact text, giving 20,958 unique passages. Because the dense encoder truncates input beyond 256 word pieces, each passage is split at sentence boundaries into chunks of at most 180 words, and the trailing sentences of a chunk (about 40 words) are repeated at the start of the next so that an answer near a boundary stays intact. This yields 23,288 chunks; the longest passage (653 words) becomes five chunks. Every chunk keeps a pointer to its parent passage.

B. Sparse Retrieval

Chunks are lower-cased and tokenised into alphanumeric terms and indexed with BM25 [3]:

$$\mathrm{BM25}(q,d) = \sum_{t \in q} \mathrm{IDF}(t)\, \frac{f(t,d)\,(k_1+1)}{f(t,d) + k_1\Big(1 - b + b\,\frac{|d|}{\mathrm{avgdl}}\Big)} \tag{1}$$

with $k_1 = 1.5$ and $b = 0.75$, where $f(t,d)$ is the frequency of term t in chunk d.

C. Dense Retrieval

Each chunk is encoded with the `all-MiniLM-L6-v2` Sentence-BERT model [4] into a 384-dimensional vector. Vectors are L2-normalised and stored in a FAISS `IndexFlatIP` index [11], so the inner product equals cosine similarity,

$$s(q,d) = \mathbf{e}_q^{\top}\mathbf{e}_d, \qquad \|\mathbf{e}_q\|_2 = \|\mathbf{e}_d\|_2 = 1, \tag{2}$$

and the search is exact.

D. Adaptive Query Analysis

A rule-based analyser does two things without any additional model. First, it classifies the question as *simple*, *comparison* or *multistep* from cue words and punctuation and sets the retrieval depth to $K = 5$, 10 or 15. Second, it writes one or two paraphrase variants of the question, so that two to three query strings are searched. Table II gives the rules. For example, "What is the speed of light?" is classified as simple and expanded to "the speed of light definition?" and "Regarding what is the speed of light?". The variants change the lexical framing seen by BM25 and the embedding seen by the dense encoder, while the cross-encoder later scores every candidate against the original question only, so a weak variant cannot distort the final order.

E. Reciprocal Rank Fusion

Every variant is searched in both indexes with 50 candidates each, giving four to six ranked lists. They are fused by position [12]:

$$\mathrm{RRF}(d) = \sum_{r \in \mathcal{R}} \frac{1}{k + \mathrm{rank}_r(d)}, \qquad k = 60. \tag{3}$$

A chunk that several lists rank reasonably high rises above a chunk that only one list ranks first, and no score calibration or weight tuning is needed.

TABLE I
POSITIONING AGAINST THE MOST CLOSELY RELATED IEEE PUBLICATIONS

Work	Approach	Limitation	This work	Our evidence
Khan *et al.* [14]	BM25 + OpenAI embeddings fused with RRF; answers generated by an LLM (BERTScore F1 93.15%)	Best setup needs proprietary models; generated answers can leave the evidence	Same BM25 + dense + RRF core with open local models; extractive answers	Hybrid Recall@10 91.50%; 96.25% with re-ranking; reader 85.38 EM / 91.74 F1
Bindle *et al.* [13]	TF-IDF/BM25 scores combined with SBERT and Word2Vec using adaptive weights	Raw scores on different scales; evaluated with BLEU and ROUGE	Rank-based fusion with no weights to tune; standard ranking metrics	Hybrid RRF: +5.35 Recall@10 and +2.21 MRR over BM25
Mageshwari and Shrijith [16]	POS filtering and FLAN-T5 query expansion with RRF (Recall@100 83.4%, nDCG@10 0.462)	A generative model runs for every query	Two to three rule-based variants; no extra model at query time	Recall@20 96.70% and NDCG@10 87.91% with multi-query fusion
Kim and Ko [17]	BERT re-ranker with multitask fine-tuning on MS MARCO	Needs labelled data and re-ranker training	Pre-trained cross-encoder applied to the top 30 candidates only	Re-ranking: +4.75 Recall@10, +13.55 MRR
Tan *et al.* [10]	LLM-generated question–answer pairs and events augment each document	LLM generation for every document at indexing time	Sentence-aware chunking with simple rules	20,958 passages indexed as 23,288 chunks; Recall@10 95.95%
Liu *et al.* [22]	One joint model for paragraph, sentence and span selection	Components cannot be swapped independently	Modular pipeline; error propagation measured directly	EM 85.38 (gold) vs. 71.00 (end to end) isolates retrieval error

TABLE II
RULES OF THE ADAPTIVE QUERY ANALYSER

Class	Trigger	K
Multistep	contains *and then, after, before, because, why did, how did, led to*, or at least two commas	15
Comparison	contains the word *compare, versus, vs, difference, better, than* or *both*	10
Simple	otherwise	5
Variant	**Rule**	
Definition	"What is X?" $\rightarrow$ "X definition?"	
Person	"Who ..." $\rightarrow$ "Which person ..."	
Topic	prefix "Regarding" to the question	

F. Cross-Encoder Re-ranking

The 30 best fused chunks are re-scored by `cross-encoder/ms-marco-MiniLM-L6-v2`, which reads the original question and each chunk jointly. Restricting the cross-encoder to 30 candidates bounds its cost while letting it correct the order produced by the first stage [2]. The K highest-scoring chunks are returned.

G. Extractive Reader

The reader receives the question and the top-ranked chunk in one sequence of at most 384 tokens and outputs start and end logits $\mathbf{s}, \mathbf{e}$ over the tokens. It is trained with the sum of the two cross-entropy losses,

$$\mathcal{L} = -\log \operatorname{softmax}(\mathbf{s})_{s^*} - \log \operatorname{softmax}(\mathbf{e})_{e^*}, \qquad (4)$$

where s^* and e^* are the gold positions. Three backbones are fine-tuned: RoBERTa-base [20] (primary, 125M parameters), BERT-base-uncased [19] (110M) and DistilBERT-base-uncased [21] (66M). The system returns the answer span with its character offsets, the evidence chunk, the relevance score, the softmax confidence of the predicted start and end, and the latency.

H. Implementation and Serving

The pipeline is implemented in Python with Hugging Face Transformers, Sentence-Transformers, FAISS and `rank_bm25`. Both indexes are built in memory at start-up; the exact inner-product index needs about 36 MB for 23,288 vectors of 384 dimensions. A FastAPI service exposes the pipeline and returns, for each question, the answer span, its character offsets in the evidence, the evidence identifier and score, the three best candidates, the detected complexity class, the generated variants, the start and end confidences and the reader latency. A web dashboard presents the results and lets a user switch between the three readers for the same question.

IV. EXPERIMENTAL SETUP

A. Dataset

SQuAD v1.1 [18] provides 87,599 training questions over 18,891 passages from 442 articles, and 10,570 development questions over 2,067 passages from 48 articles. Passages average 117 words. The retrieval corpus is the union of both splits (20,958 passages).

B. Metrics

Retrieval is scored on 2,000 development questions, where the relevant item is the passage the question was written for and retrieved chunks are mapped to their parent passage. For

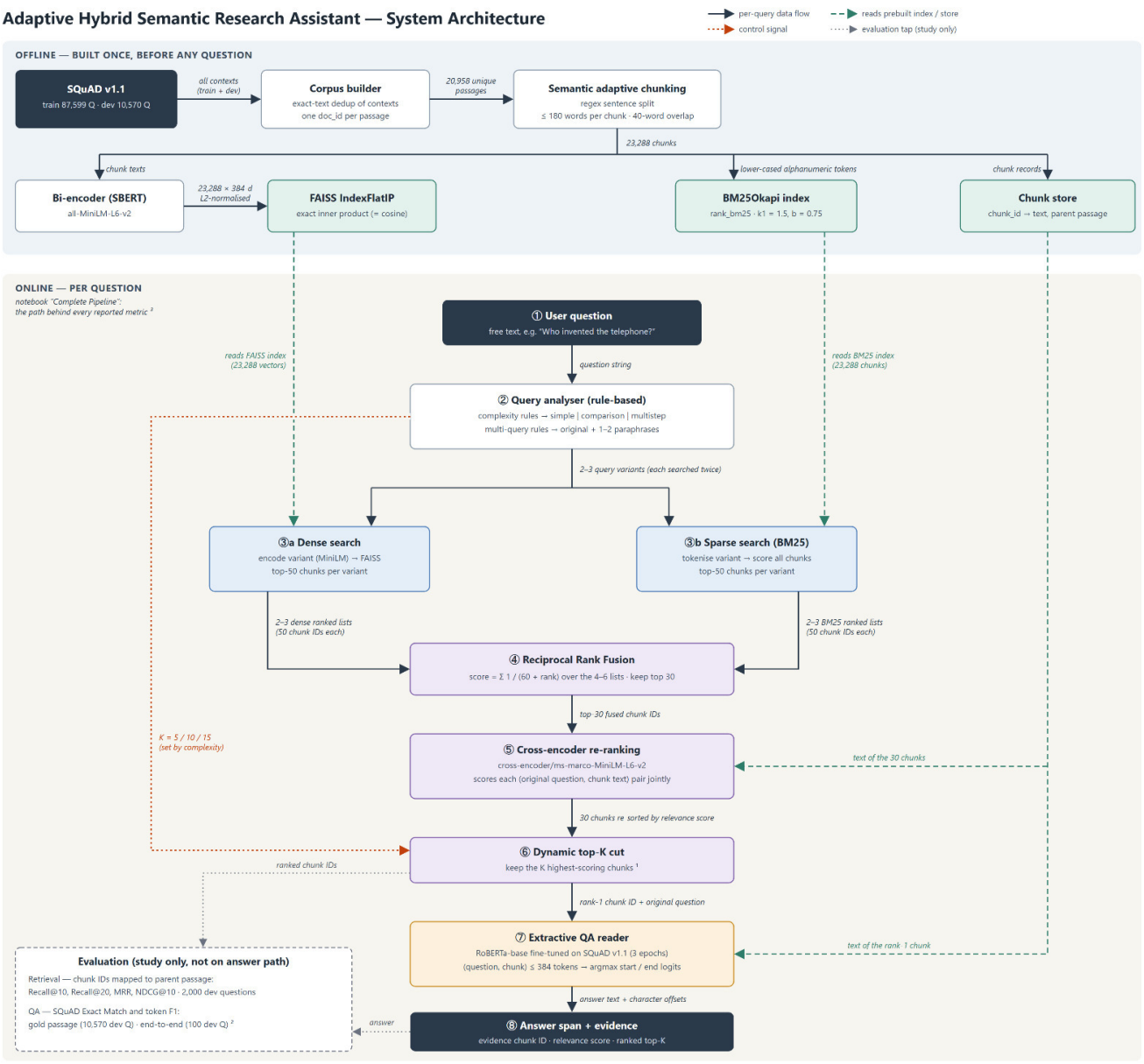


Fig. 1. System architecture. The offline pipeline (top) builds the dense index, the BM25 index and the chunk store. The online pipeline (bottom) analyses the question, searches both indexes for every query variant, fuses and re-ranks the candidates, and extracts the answer span from the top-ranked chunk.

a question set Q with gold passage g_q and ranked list π_q we report

$$\text{Recall@}K = \frac{1}{|Q|} \sum_{q \in Q} \mathbf{1}\left[g_q \in \pi_q^{(1:K)} \right], \tag{5}$$

$$\text{MRR} = \frac{1}{|Q|} \sum_{q \in Q} \frac{1}{\text{rank}_q(g_q)}, \tag{6}$$

$$\text{NDCG@}K = \frac{1}{|Q|} \sum_{q \in Q} \frac{\text{DCG}_q@K}{\text{IDCG}_q@K}, \tag{7}$$

where $\text{DCG@}K = \sum_{i=1}^{K} \text{rel}_i / \log_2(i+1)$, with $K \in \{10, 20\}$ for recall and $K = 10$ for NDCG. Recall@K bounds what any reader can achieve, while MRR and NDCG reward placing the gold passage early, which matters because the reader consumes the top-ranked chunk. Reading is scored with the official SQuAD Exact Match (EM) and token-level F1, taking the maximum over the reference answers of each question:

$$\text{F1} = \frac{2PR}{P+R}, \quad P = \frac{|\hat{a} \cap a^*|}{|\hat{a}|}, \quad R = \frac{|\hat{a} \cap a^*|}{|a^*|}, \tag{8}$$

where $\hat{a}$ and a^* are the token multisets of the normalised prediction and reference. Readers are evaluated in two settings: with the gold passage, which isolates reading quality, and end to end, where the complete retrieval pipeline supplies the passage.

TABLE III
CONFIGURATION

Setting	Value
Chunk size / overlap	180 / 40 words
Dense encoder	all-MiniLM-L6-v2 (384-d)
Candidates per list	50
RRF constant k	60
Re-ranked candidates	30
Dynamic top-K	5 / 10 / 15
Max sequence / stride	384 / 128 tokens
Epochs	3
Batch size	16 × 2 accumulation (effective 32)
Learning rate	1.5e-5 (RoBERTa), 3e-5 (BERT), 5e-5 (DistilBERT)
Schedule	linear, 6% warm-up
Weight decay / clipping	0.01 / 1.0
Precision	FP16

TABLE IV
RETRIEVAL ABLATION ON 2,000 SQUAD DEVELOPMENT QUESTIONS (%). S: SPARSE, D: DENSE, CE: CROSS-ENCODER, CH: CHUNKING, MQ: MULTI-QUERY

Configuration	S	D	CE	Ch	MQ	R@10	R@20	MRR	NDCG
BM25 only	✓					86.15	89.60	69.49	73.39
Dense only		✓				86.00	90.55	63.83	68.97
Hybrid (RRF)	✓	✓				91.50	94.60	71.70	76.37
+ Re-ranking	✓	✓	✓			**96.25**	96.65	**85.25**	**87.97**
+ Adaptive chunking	✓	✓		✓		91.35	94.40	71.56	75.64
+ Multi-query	✓	✓	✓		✓	**96.25**	**96.70**	85.19	87.91
Complete pipeline	✓	✓	✓	✓	✓	95.95	96.45	85.20	87.31

C. Training Configuration

Table III lists the configuration. Each reader is fine-tuned for three epochs on the full training set with mixed precision on one NVIDIA GeForce RTX 5060 Laptop GPU (8 GB), using PyTorch 2.12 with CUDA 12.8. Long passages are handled with a sliding window of 384 tokens and stride 128.

V. RESULTS

A. Retrieval Ablation

Table IV and Fig. 2 report the seven configurations. BM25 and dense retrieval are individually close (86.15% and 86.00% Recall@10), yet their fusion reaches 91.50%, which confirms that the two make different mistakes. Adding the cross-encoder gives the largest single gain: Recall@10 rises to 96.25% and MRR from 71.70% to 85.25%, a gain of 13.55 points. Multi-query fusion with re-ranking reaches the best Recall@20 (96.70%).

In terms of errors, BM25 fails to place the gold passage in the top ten for 13.85% of questions. The re-ranked hybrid fails for 3.75%, so the stack removes 73% of the misses of the baseline (Fig. 3). The complete pipeline, which also returns chunk-level evidence, keeps Recall@10 at 95.95% and improves on BM25 by 9.80 points of Recall@10, 15.71 points of MRR and 13.92 points of NDCG@10. Table V summarises the gain of each step. The cost is modest: on the evaluation hardware, hybrid retrieval takes about 66 ms per question and the re-ranked hybrid about 150 ms.

TABLE V
GAIN CONTRIBUTED BY EACH RETRIEVAL STEP (PERCENTAGE POINTS)

Step	ΔR@10	ΔMRR	ΔNDCG@10
BM25 → Hybrid (RRF)	+5.35	+2.21	+2.98
Dense → Hybrid (RRF)	+5.50	+7.87	+7.40
Hybrid → + Re-ranking	+4.75	+13.55	+11.60
BM25 → Complete pipeline	**+9.80**	**+15.71**	**+13.92**

TABLE VI
READER COMPARISON ON SQUAD V1.1. GOLD: CORRECT PASSAGE SUPPLIED (10,570 QUESTIONS). E2E: PASSAGE SUPPLIED BY THE COMPLETE RETRIEVAL PIPELINE (100 QUESTIONS)

		Gold		E2E		Latency	Train
Reader	Params	EM	F1	EM	F1	(ms)	(h)
RoBERTa-base	125M	**85.38**	**91.74**	**71.00**	**72.34**	8.3	2.45
BERT-base	110M	80.18	87.85	62.00	67.54	7.9	2.33
DistilBERT	66M	76.98	85.52	60.00	62.41	**4.3**	**1.18**

B. Reader Comparison

Table VI compares the three readers. With the gold passage supplied, RoBERTa-base reaches 85.38 EM and 91.74 F1 on all 10,570 development questions, ahead of BERT-base by 5.20 EM and 3.89 F1 and ahead of DistilBERT by 8.40 EM and 6.22 F1. The ordering agrees with the pre-training improvements reported for RoBERTa [20] and with the compression trade-off reported for distilled models [21], [24].

Fig. 4 shows the training curves. All three models converge smoothly within three epochs. RoBERTa-base attains the lowest evaluation loss (0.857 after the second epoch); the checkpoint with the lowest evaluation loss is kept for each model. Table VII gives the training statistics: the two 12-layer models process about 30 examples per second and finish in under 2.5 hours, and the 6-layer DistilBERT runs at twice that rate.

C. Accuracy and Efficiency

Fig. 5 relates accuracy to cost. DistilBERT has 53% of the parameters of RoBERTa-base, answers in 4.3 ms instead of 8.3 ms and trains in under half the time, while retaining 93.2% of its F1. RoBERTa-base costs only 0.4 ms more per question than BERT-base and gains 3.89 F1, which makes it the clear choice when accuracy matters and DistilBERT the choice under a latency budget. All three readers answer in under 10 ms.

D. End-to-End Evaluation

In the end-to-end setting the reader sees only the top-ranked chunk returned by the complete pipeline. RoBERTa-base answers 71.0% of questions exactly (72.34 F1), retaining 83% of its gold-passage EM with no oracle (Fig. 6). Its margin over BERT-base widens from 5.20 EM with gold passages to 9.00 EM end to end, so the stronger reader is also the more robust one when the context is retrieved. The remaining gap between the two settings is the retrieval-to-reader error propagation described in [22]; it is bounded by

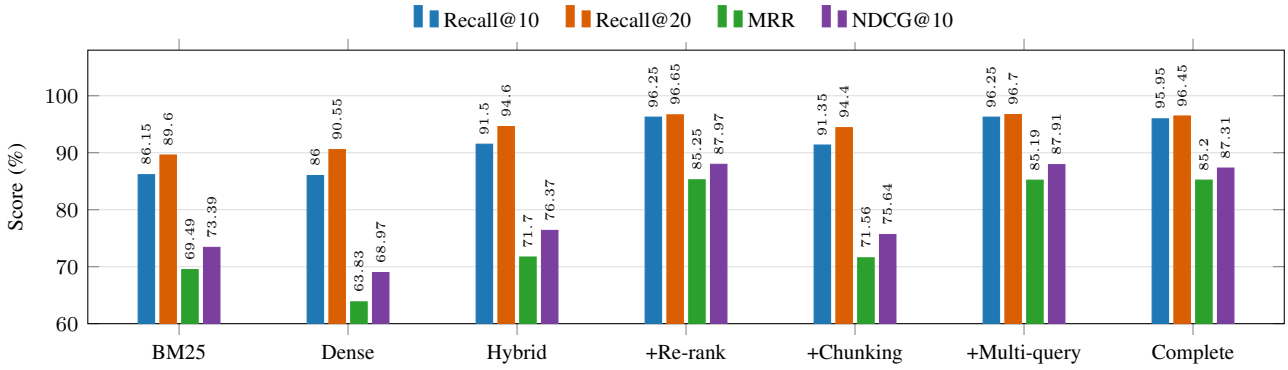


Fig. 2. Retrieval quality of the seven configurations. Fusion lifts recall; the cross-encoder lifts the rank-sensitive metrics (MRR, NDCG@10) by more than 11 points.

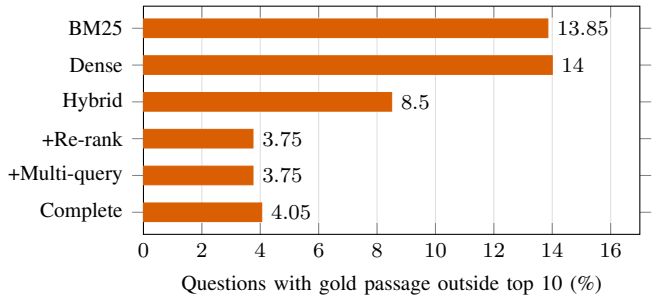


Fig. 3. Retrieval miss rate at rank 10. Each stage removes misses; re-ranking brings the rate from 13.85% to 3.75%.

TABLE VII
TRAINING STATISTICS (3 EPOCHS, 87,599 QUESTIONS)

Reader	Steps	Ex./s	Time (h)	Train loss	Best eval
RoBERTa-base	8,304	30.1	2.45	0.938	**0.857**
BERT-base	8,301	31.6	2.33	0.980	1.003
DistilBERT	8,301	62.7	1.18	1.046	1.100

rank-1 accuracy, because the reader cannot extract an answer from a chunk that does not contain it.

E. Comparison with Published Results

Table VIII sets the readers against published SQuAD v1.1 results. On the same benchmark and metric, the RoBERTa-base reader:

- exceeds the original logistic-regression baseline [18] by 40.7 F1;
- exceeds human performance [18] by 8.4 EM and 4.9 F1;
- exceeds the published BERT-base result [19] by 4.6 EM and 3.2 F1, and the published DistilBERT result [21] by 7.7 EM and 5.9 F1;
- exceeds the published single-model BERT-large result [19] by 1.3 EM and 0.8 F1 with about a third of the parameters;
- is within 0.5 EM and 0.1 F1 of the BERT-large ensemble [19].

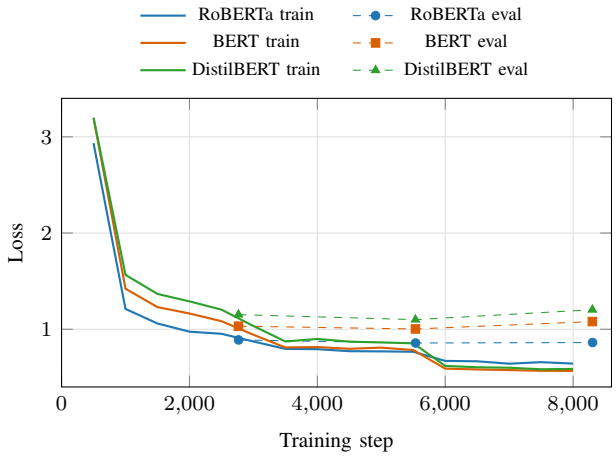


Fig. 4. Training loss (solid) and per-epoch evaluation loss (dashed) of the three readers.

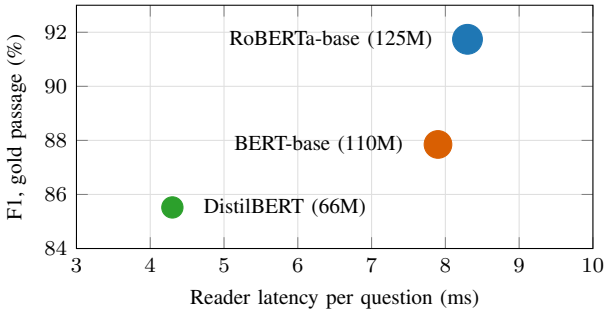


Fig. 5. Accuracy against reader latency. Marker size is proportional to parameter count.

Our own BERT-base and DistilBERT readers land within 0.7 F1 of their published scores, which confirms that the training pipeline is sound.

For orientation on the end-to-end task, Karpukhin *et al.* [8] report open-domain EM on SQuAD questions between 20.2 and 56.5 for systems that search all of Wikipedia, and a top-20 retrieval accuracy of 68.8% for BM25 there. Those systems search a corpus of about 21 million passages whereas ours

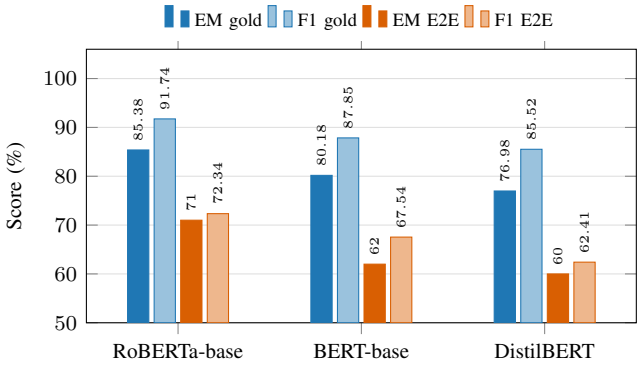


Fig. 6. Gold-passage and end-to-end accuracy of the three readers.

TABLE VIII
COMPARISON WITH PUBLISHED SQUAD V1.1 RESULTS (DEVELOPMENT SET UNLESS NOTED)

System	EM	F1
Logistic regression [18]	40.0	51.0
Human performance (test) [18]	77.0	86.8
DistilBERT [21]	77.7	85.8
BERT-base, single [19]	80.8	88.5
BERT-large, single [19]	84.1	90.9
BERT-large, ensemble [19]	85.8	91.8
Ours: DistilBERT	76.98	85.52
Ours: BERT-base	80.18	87.85
Ours: RoBERTa-base	**85.38**	**91.74**

searches the 20,958 SQuAD passages, so the figures are not directly comparable; they show the regime in which our 71.0 EM and 96.70% Recall@20 sit. The same caution applies to the retrieval scores of [16] (Recall@100 83.4%, nDCG@10 0.462), which were measured on a different collection.

F. Qualitative Examples

Table IX lists answers returned by the deployed demonstration, which serves the hybrid retriever with the RoBERTa-base reader. Each answer is a span of the retrieved passage and is returned with the span confidence and the reader latency.

VI. Novelty and Discussion

The individual components are established; the novelty lies in how they are combined, adapted and measured.

Adaptive retrieval without extra models. Related systems adapt retrieval with learned weights [13] or with a generative model that runs for every query [16]. Here, complexity estimation and multi-query expansion are pure rules, so adaptation adds only extra index look-ups.

Open and local. The best configuration of a comparable hybrid system relies on proprietary embeddings and a proprietary generator [14]. Every model here is open and the full study, including three fine-tuning runs of 1.2 to 2.5 hours, runs on one laptop GPU.

Faithful by construction. The answer is a span of the retrieved chunk, returned with the chunk, its score, the character offsets and a confidence. This addresses the faithfulness risk discussed in [5] structurally instead of by post-hoc checking, and complements chunk-level grounding methods such as [25].

TABLE IX
EXAMPLE ANSWERS FROM THE DEMONSTRATION (ROBERTA-BASE READER)

Question	Answer	Conf.	ms
Which NFL team represented the AFC at Super Bowl 50?	Denver Broncos	0.89	8.1
In what year was Nikola Tesla born?	1856	0.99	8.7
What is Southern California often abbreviated as?	SoCal	0.99	6.9
When did the 1973 oil crisis begin?	October 1973	0.97	7.3
What project put the first Americans into space?	Project Mercury	0.94	8.1
What are Ctenophora commonly known as?	comb jellies	0.96	11.8

Stage-by-stage evidence. The ablation isolates what each stage buys: fusion adds 5.35 points of Recall@10 over BM25, and re-ranking a further 4.75 points of Recall@10 and 13.55 points of MRR. It also reports honestly where a stage does not help on this benchmark: SQuAD passages average 117 words, so chunking leaves recall almost unchanged, although it supplies shorter, more focused evidence and matters more for longer documents [6], [10].

Locating the bottleneck. Reader comparisons usually evaluate on given passages [23]. Evaluating the same readers in both settings shows that the reader is already strong and that further gains should come from rank-1 retrieval accuracy.

Deployable artefact. The system is served through a FastAPI endpoint and a static web dashboard that renders every reported number from the committed result files.

VII. Limitations and Future Work

The reader is single-hop and always returns a span, so it cannot answer questions that need several passages or decline unanswerable ones; training on SQuAD 2.0 would address the latter. The query variants are rule-based, and a small generative paraphraser could increase their diversity. The bi-encoder is used off the shelf; fine-tuning it with hard negatives [8], [9] is the most direct route to higher rank-1 accuracy and therefore to higher end-to-end accuracy. Evaluating on a larger open-domain corpus is a further step.

VIII. Conclusion

This paper presented a retrieval-first, evidence-grounded QA system that combines sparse and dense retrieval, rank fusion, adaptive query analysis, cross-encoder re-ranking and extractive reading. On SQuAD v1.1 it reaches 96.25% Recall@10 and 85.25% MRR, removing 73% of the retrieval misses of BM25, and its RoBERTa-base reader reaches 85.38 EM and 91.74 F1, above human performance and above the published BERT-base, DistilBERT and single-model BERT-large results. End to end it answers 71.0% of questions exactly, with evidence, confidence and latency attached to every answer. The system achieves this with open models on a single laptop

GPU, which makes grounded, verifiable question answering practical on modest hardware.

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